STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-FUNCTOR-SET-OF-MONOS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4183-4207 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4209-4230, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 33 of queue_sdga.json (pb=819, stmt 4183-4207, proof 4209-4230).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let R be a set and for each $r \in R$ let an injective map $\mathcal{M}_r \to \mathcal{M}'_r$ of acyclic differential graded $\mathcal{A}$-modules be given. There exists a functor $M : Mod(\mathcal{A}, d) \to Mod(\mathcal{A}, d)$ and a natural transformation $j : id \to M$ such that*

(1) *$j_{\mathcal{M}} : \mathcal{M} \to M(\mathcal{M})$ is injective and a quasi-isomorphism,*
(2) *for every solid diagram*

$$\begin{array}{ccc} \mathcal{M}_r & \longrightarrow & \mathcal{M} \\ \downarrow & & \downarrow{\scriptstyle j_{\mathcal{M}}} \\ \mathcal{M}'_r & \dashrightarrow & M(\mathcal{M}) \end{array}$$

a dotted arrow exists in $Mod(\mathcal{A}, d)$ making the diagram commute.

Proof. We define $M(\mathcal{M})$ as the pushout in the following diagram

$$\begin{array}{ccc} \bigoplus_{(r,\varphi)} \mathcal{M}_r & \longrightarrow & \mathcal{M} \\ \downarrow & & \downarrow \\ \bigoplus_{(r,\varphi)} \mathcal{M}'_r & \longrightarrow & M(\mathcal{M}) \end{array}$$

where the direct sum is over all pairs (r, φ) with $r \in R$ and $\varphi \in \mathrm{Hom}_{Mod(\mathcal{A},\mathrm{d})}(\mathcal{M}_r, \mathcal{M})$. Since the pushout of an injective map is injective, we see that $\mathcal{M} \to M(\mathcal{M})$ is injective. Since the cokernel of the left vertical arrow is acyclic, we see that the

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(isomorphic) cokernel of $\mathcal{M} \to M(\mathcal{M})$ is acyclic, hence $\mathcal{M} \to M(\mathcal{M})$ is a quasi-isomorphism. Property (2) holds by construction. We omit the verification that this procedure can be turned into a functor. $\square$